Quantitative Finance is Information Theory

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The Initial Analogy: *Implied Volatility ≈ Entropy*

The Black Scholes Option Pricing Formula calculates something akin to entropy or average information of underlying assets. Options are overpriced 85% of the time—that is, in 15 out of 100 periods priced by an option, the underlying asset will land outside of the range prescribed by the option. Said a third way, if I sell weekly options for the next 100 weeks at theoretical value and delta hedge (perfectly, for argument's sake), I will be profitable in 85 of weeks. However—still in our ideal world—I will end the 100 weeks with totally flat profit. All of the savings made in the regular cases will be wiped out by disproportionately unexpected price movements in the rarer cases. For options, the edge cases disproportionally (relative to how many such edge scenarios there actually are) add to the price you need to cover the uncertainty.

This is reminiscent of variable length encodings of information in information theory: entropy is the average number of bits needed to encode a word or unit in some system or language. If the units of encoding are letters, then most of the letters in a larger stream of text will be encoded with fewer bits than entropy, but for some letters like 'q', we will need more than average. The same goes for unicode characters—obscure emojis will cost us compared to 'e'—and any other variable length encoding system such as Huffman.

The analogy here is not by any means 1-1 (at least not yet), but in both scenarios, we have a quantification of surprise coming from a stream of information. It is a compression or encoding of average/expected uncertainty into a single quantity. When encoding messages, we want to be as efficient as possible—surprising messages will cost us more. When covering a short option position, we want to hedge away all of our risk—unexpected price moves will cost us more. Another vital requirement in both scenarios is the assumption of an ergodic stream of information. Entropy is only exact when the source is ergodic (different messages are sent with constant probabilities) and Black Scholes only works for stocks with constant volatility. Without this assumption, then neither of these *at the probabilistic limit* values can accurately describe what they seek to.

What is the Black Scholes price? It is the present value of all of the net payments you will make to maintain zero delta risk until expiration of the option. It is the net cost to wrangle the beast of the unpredictable market, to meet every day's price movements and lose nothing. You need to properly represent (be hedged at) every moment. This then sounds just like the binary encoding of a stream of information. Each day, uncertainty needs to be covered by your portfolio. When you design an encoding system for a language (or any other context), you need to be able to meet every word (unit of encoding) with a representation. You want to spend as little as possible to hedge your option. You want to use as few bits as possible to encode each message. But you are unhedged if something needs encoding and you do not represent it. You lose information (and money!) when you are unhedged to new uncertainty.

Both of these values are doing statistical mechanics on their respective contexts. They measure macroscopic characteristics about underlying probabilistic systems. They abstract away the unimportant instantiation of *what word actually comes next* or *what the stock does today* and instead talk about how, on average, you'd compress their systems' uncertainty into a value or how you'd represent events you observe.

Since both are about representing uncertainty in widely used systems by looking at their macro statistics, they can tell us fundamental properties of the systems themselves. That is, to encode a language you need artifacts of that language *or* an idealized ergodic equation describing its artifacts. Then you fit your encoding and out pops this entropy statistic—the average length of each unit's code word, the average information in each unit. To price an option, you either need stock price data *or* an idealized ergodic equation describing how its price changes. Then you price your option and out pops an implied volatility. **Thus we see that *implied volatility* (not the option price) is really the kin to entropy.** Specifically, it is *the square root of* implied volatility, or implied variance (we will see this later).

Entropy tells you how potent or redundant a language is. Implied volatility tells you how shifty or stable a stock is. These are both extremely valuable pieces of information that arise from the statistics of a system that you use to represent its natural expression.

- We start with language, we impose an optimal encoding system upon it, and the statistics of that encoding tell us something about the language.
- We start with a stock, we create and optimally hedge a derivative asset atop it to encode things about where it may go, and from the statistics of this derivative asset (really the portfolio that is isomorphic to it) we glean useful information about the stock.

Stock prices change as a function of humans trading them. Language is expressed naturally by humans. The insights we glean about these complex sources of expression come from our creation of mathematically optimal appendages to them.

A Foundational Equivalence: *Kelly Wealth = KL Divergence = Information Edge*

The connections between information theory and finance have definite origins and can explain why Black Scholes price resembles as an informational metric. Information theory and quantitative finance are principally interested in understanding uncertainty, making use of probability and statistics to compress human behavior. Claude Shannon, working at Bell Telephone Laboratories, codified this generalized study of information into a powerful theory composed of several equations. In his view, information is *surprise*. It is deviance from pattern. If I *know* what someone is going to say, I need not even listen to them. If I know exactly what a stock will do, I need not worry about hedging uncertainty, *I can go all in*. He was working on the endless stream of problems that came from making AT&T's national communications systems as efficient as possible. He wanted information to be compressed and transmitted safely, quickly, and in whole! The foundational quantity is Entropy. Suppose I am just looking at the entropy of a system with binary events, like a coin flip. It tells us the average information (surprise) that each event gives us. Here is the equation:

$$H = -P_1 \log_2(P_1) - P_0 \log_2(P_0),$$

where P_1 is the probability of a 1 or *heads* or *the event happening* and P_0 is that of a 0 or *tails* or *the event not happening*. What is the average information gained in one coin flip? Well P_1 (heads) and P_0 (tails) are both $50\,\%$, so H becomes:

$$\begin{aligned} H &= -0.5\log_2(0.5) - 0.5\log_2(0.5) \\ &= -0.5(-1.0) - 0.5(-1.0) \\ &= 0.5 - -0.5 \\ &= 0.5 + 0.5 \\ &= 1.00 \end{aligned}$$

This makes sense! If some system has perfectly symmetrical randomness, then we learn one bit, or a full binary choice, by sampling it. Alternatively if we wanted to convey to someone what the results of a coin flip are, I cannot rely on any kind of pattern, I cannot say something like ‘normal result’; I need to just fully represent whether it was *heads **or** tails*. If we have a totally random system, we are surprised by anything, because we can *expect* nothing! Suppose, though, that heads was weighted at 0.6 and tails at 0.4. We’d have:

$$\begin{aligned} H &= -0.6\log_2(0.6) - 0.4\log_2(0.4) \\ &= -0.6(-0.74) - 0.4(-1.32) \\ &= 0.44 - -0.53 \\ &= 0.44 + 0.53 \\ &= 0.97 \end{aligned}$$

Here we go! Now the equation is telling us something nontrivial. In this system, one of the outcomes is more likely, so, even though the rarer case surprises us, on average, we’re less surprised.

Let’s examine the intuition of this equation. First, let’s separate each term so that the negative sign is applied to the log_2. The $-log_2$ of a number less than one gets exponentially higher the closer the number is to 0.0. In this way, as events are less likely, their $-log_2$ shoots up exponentially. We could then say that $-log_2$ is a measure of surprise or rarity in that event. We are *very* surprised when something unlikely happens. But the *-log* term is applied to every possible event, so if we want to think about the weighted *average* surprise for *any* given outcome, we want to scale this surprise.

It only seems sensible then that *the surprise induced by the event happening* should be exactly proportional to *the likelihood of that event happening*. We see this perfect balance in our example. Since tails, with probability 0.4, is less likely and therefore more surprising than heads, its $-log_2$ value is 1.32 (relatively much higher than heads’s at 0.74). But this surprise only happens $40\,\%$ of the time, so we scale it accordingly. Heads is less surprising when it comes, but comes more often.

This powerful measurement of information can be applied masterfully to any system with a finite amount of possible outcomes, such as the choice of a word in a person’s speech. If there are 50,000 possible words, the entropy equation will have 50,000 terms (as opposed to our two). The general equation is:

$$H = -\Sigma_i^n P_i \log_n(P_i)$$

In other words, a summation of the *probability-weighted negative log probability* of each event happening. If we think about English letters, as opposed to words, we can imagine what some of the terms of our equation may look like. For a common letter like ‘e’, we will have a low $-log$ that is scaled by a high *P*. For a less common letter like ‘q’, we will have a very high $-log$ value that is scaled back down by its low chance (*P*) of occurring. You can apply this reasoning everywhere.

Expanding on entropy, we can calculate additional powerful statistics about certain distributions or systems. Beyond just measuring the average information in one system, we may want to measure how similar two different systems/sources/languages/distributions are to each other. Remember in each term of *H*, we have the $-log_n$ measuring the *surprise* induced by the observation of some outcome and then the 'true' probability of that outcome actually being observed. When we are looking at one system, the *P* values that go into the $-log_n$ term and the outer scaling term are the same. But if we wanted to measure how similar, say, the informational content of our heads-weighted coin was relative to a fair coin, what would we do? We might just use the fair coin's *P*s for one of the values and the weighted coin's for the other. I like to think of entropy as scaling *actual surprise* by *expected* likelihood of surprise. The outer *P* term is a normalizing baseline, what we *expect*. We use the real probability to scale whatever surprise we calculated. So if we use the fair coin's distribution for the raw scaling *P* terms and the weighted coin's distribution as that which informs the $-log_n$ surprise terms, then we can bring in the *surprise* from the system-under-observation into the *expected* context of the baseline system.

So we flip our weighted coin many times and observe the statistics, we will get 0.6 for heads and 0.4 for tails. When we calculate this new composite measure of the *weighted coin's surprise* in terms of the *fair coin's expectation*, we get a value of: $-0.5(-0.74) - 0.5(-1.32) = 1.03$. Recognize that if we used the fair coin's stats for both, we would get 1.00 again. Whereas the self-scaling in *H* made the system perfectly sensible, here we see that it does something a bit different. It measures entropy *across* the two distributions. It measures the average surprise if we *expect* a fair coin and *observe* a weighted one. Thus, it is commonly called the *cross entropy* of the weighted coin with respect to the fair coin.

Cross entropy is extremely ubiquitous in statistics and data science, most impactfully as a measure of how accurate a machine learning model is. If we *expect* a neural network to produce a certain set of outputs (a baseline), then we can use cross entropy to measure how much its *observed* behavior surprises us. We seek to minimize cross entropy and produce a model that behaves exactly as we expect!

While our fair coin example got our feet wet with entropy and cross entropy, I need another scenario to massage out the nuance between cross entropy and another very important value in statistics. The way it has been painted, the cross entropy measures how one distribution diverges from another. However it does so somewhat imprecisely. Indeed, cross entropy measures some aspect of the divergence of one distribution from another. But it does not isolate this divergence. It tells us *the average surprise of expecting distribution P and observing distribution Q*. But what if *P* itself has a very high entropy? In that case, we'd be looking at the surprise relative to an already surprising system. This somewhat muddies the water of what we initially thought we were observing. To observe the surprise of any system *relative to any other baseline* system, we need to somehow factor out the actual entropy of the baseline system. Now that we have the intuition, the math is rather straightforward.

Suppose instead of our baseline fair coin, we have a baseline coin, *P,* with the probability of heads being 0.8 and tails being 0.2. First, we'll measure both of our entropies:

$$H_P \approx 0.72$$
$$H_Q \approx 0.97$$

Now the cross entropy of *Q* with respect to *P*:

$$CE_{P,Q} = 0.85$$

Pausing for a moment to ponder, we can see why we need something else to isolate *just* how *Q* surprises us when we are expecting *P*. Cross entropy now reveals itself as not just measuring divergence of *Q* from *P*, but their combined divergence from the baseline of 0.00. Now, we do not want to confuse ourselves into thinking that we have simple addition of entropies here (try P with respect to Q!), but we do want to see that when these two distributions interact, we have more complicated interactions contributing to *net* surprise. The problem is that $CE_{P,Q}$ alone carries with it the informational content inherent in P. What we want is $CE_{P,Q}$ but without any influence of P; in other words, we want to *factor P*'s *surprise out* of the equation to isolate just the *additional* surprise of Q. Since we are dealing with *log* probabilities, factoring is as simple as subtracting P's entropy from the equation! It is not directly there, but it is hidden in the shared information of P and Q. This divergence value we are gesturing at is known as the *Kullback-Leibler (KL) Divergence,* defined precisely as:

$$KL_{P||Q} = H_{P,Q} - H_P$$

Or, expanded:

$$KL_{P||Q} = -P_1 \log_2 \frac{P_1}{Q_1} - P_0 \log_2 \frac{P_0}{Q_0}$$

Or, generalized:

$$KL_{P||Q} = -\Sigma_i^n P_i \log_2 \frac{P_i}{Q_i}$$

(The fact that P is numerator looks deceiving, but that is how the math works out when you simplify! Breadcrumbs: we subtract H_P which has negative terms, thus adding them to our equation…)

This quantity precisely measures the divergence—in terms of average information and surprise—of the distribution Q with respect to P. This is also massively useful in training machine learning models, specifically in later stages where we actually want to *minimize* KL Divergence to ensure that a model's behavior does not deviate too far from a baseline. It can also be useful in other contexts like gambling (and finance!). How?

Note: this is not a normative piece about ethics and I do not encourage betting with illegally privileged information. But imagine you come upon someone offering a coin-flip gambling game. The dealer takes your guess for what you think the coin will be when he flips it and you bet a certain amount of money on the flip. He, being the enterprising gambling clerk that he is, advertises that if you win on heads, he'll give you back \$1.67 for every \$1.00 you bet and, for tails, \$2.5 for every \$1.00. That means for heads, your profit odds are $\frac{0.67}{1}$ and for tails $\frac{1.5}{1}$. Working out the numbers (we'll see precisely how below), he is offering payouts for a coin that lands heads $60\,\%$ of the time and tails $40\,\%$ of the time. This gets players very excited. Since they do not have all day to sit around and count outcomes, they assume he is actually playing with a weighted coin. But you, who know this trickster well, know that he is just using a normal coin with even odds! There is an opportunity here for you. The surprise being *compensated by the market* is different from that which actually exists in the system governing the payouts. There is extra information in the system! In theory, you theorize, there is (on average) a certain amount of information to be extracted on every bet you place. Now of course you'll lose sometimes, but we are talking about average. Clearly you'll have to plan your bets accordingly so as to not wipe yourself out on a first unlucky bet…what is the best strategy?

First, let's calculate the *best case scenario* of informational edge you can extract on each bet. How do we measure this…well with KL Divergence! The distributions we are interested in are that of the fair coin and that of the 60/40 odds being offered by the flipper. When *I expect* even odds characteristic of a fair coin, how much extra information (payout) is observed? Well this brings back our old result:

$$KL_{P||Q} = H_{P,Q} - H_P$$
$$KL_{P||Q} = 1.029 - 1.00$$
$$KL_{P||Q} = 0.029$$

While we do not yet have units calibrated, we know that the maximum average advantage we can extract from our privileged information is proportional to this number. Now how do we determine the optimal bet size to actually achieve this without bankrupting ourselves?

Enter John Kelly!

Let's briefly reflect on what exactly this information or payoff edge is and where it comes from. In reality, we have a fair coin that lands heads 50 % of the time and tails 50 % of the time. To bet on this coin, you *should* get \$1 profit for every \$1 you wager if you are correct. The probabilities (50 %) and the payoffs or odds (1 to 1) have a direct correspondence to each other. In this binary scenario, to go from probabilities to odds, you simply subtract the probability of YES from 1.0, then divide that number by the probability. Explicitly we have:

$$Payoff = \frac{1 - Probability}{Probability}$$

But for the weighted coin in this scenario, I started with the payoff and then derived the probabilities from it. Here is how:

$$Probability = \frac{1}{1 + Payoff}$$

There we go! This is a fundamental identity and it gives us actionable insight. We knew the *payoff* of the coin being offered and we knew the *probability* that governed the coin. Now we know the *probability* implied by the bet odds and we know what *payoff* we should actually be offered for a fair coin. It is vital to be able to put these into like terms so that the discrepancy becomes obvious in context.

Once again, when I bet heads and am correct, I get \$0.67 in *profit* per dollar bet; when I bet tails and am correct, I get \$1.5 in profit per dollar bet. What I *should* get for these scenarios is \$1 and \$1, respectively. When I win in this offered game on *heads*, I get the short end of the stick. But when I am correct on *tails*, I get a huge payoff. Now let's imagine the four possible scenarios in our game:

$$Profit(H, H) = \$0.67 \qquad \checkmark$$
$$Profit(H, T) = -\$1.00 \qquad \times$$
$$Profit(T, T) = \$1.5 \qquad \checkmark$$
$$Profit(T, H) = -\$1.00 \qquad \times$$

The first corresponds to the profit I get when if I bet \$1.00 on *Heads* and the coin lands on *Heads*. Let's quickly see our expected profit, in absolute terms again, for just betting *heads* every time:

$$\mathbb{E}_{Profit_H} = 0.5 \times \$0.67 - \$0.5$$

$$\mathbb{E}_{Profit_H} = -\$0.167$$

Yikes. What about all tails?

$$\mathbb{E}_{Profit_T} = 0.5 \times \$1.5 - \$0.5$$
$$\mathbb{E}_{Profit_T} = \$0.25$$

It seems like betting tails is the right move here. This makes sense because we are getting paid as if it is rarer than it truly is.

Now that we have detected the presence of our information edge, we need to extract it. John Kelly shows us how. We need to know the fraction of our wealth that we should place on each bet in this scenario to maximize our long term wealth growth. He models our wealth growth as follows:

$$G_f = p \ln(1 + bf) + q \ln(1 - f)$$

Where p is the probability of *heads*, q the probability of *tails*, and b the payoff we get for betting correctly on heads. We use natural logarithm (*ln*) here because this equation is meant to represent long term compound wealth growth and e is conventionally better fit for this value (we can also always convert back to any log base by dividing by $\ln 2$, a constant). This equation looks mighty familiar—a probability factor scaling a *log* term that contains something relative to a payoff (which implies a probability). Which equation might this resemble? It is not entropy, H, because b and p represent different distributions. It is either *cross entropy* of the b-implied distribution with respect to p or the KL Divergence; does this equation factor out the entropy of p? Let's start with KL. In that case, $1 + bf$ should be analogous to $\frac{p}{q}$. Is it?

$$\frac{p}{q} = \frac{1-q}{q}$$

Well, remembering our earlier identity between *payoff* and *implied probability*, we know that *payoff* is equal to $\frac{1 - Probability}{Probability}$! In this scenario, we need to be careful about what p is. b tells us what the *implied market probability, M*, is. Here, for heads (for a b_H), $M_H = 0.6$. Before we can fully unify, we need to know what that optimal f is! We have the equation which tells us our expected wealth growth per bet in logarithmic terms of betting f of our wealth on *heads*. When we maximize G_f we get the following:

$$f_* = p - \frac{q}{b}$$

Now we can put all of our pieces together.

$$1 + b_H f_* = 1 + b_H(p - \frac{q}{b_H})$$

$$= 1 + b_H p - q$$

$$= (1 - q) + b_H p$$

$$= p + b_H p$$

$$= p(1 + b_H)$$

Recalling that for an arbitrary payoff, we have $P = \frac{1}{1 + Payoff}$, so $1 + Payoff = \frac{1}{P}$:

$$= p(\frac{1}{M_H})$$

$$1 + b_H f_* = \frac{p}{M_H}$$

This is huge! Why? Because M_H is the market-implied probability of *heads* landing—in other words, it is the market distribution's alternative to p itself. Recall that in $KL_{P||Q}$, the amount by which Q diverges from P, we have, in the first log term, $\frac{P}{Q}$. Here, big P is p, the true probability of *heads*. Big Q is M_H, the alternative probability for *heads*. So we can first rewrite the following:

$$G_{f_H*} = p \ln \frac{p}{M_H} + q \ln(1 - f_{H*})$$

Now, following a very similar derivation as for $1 + b_H f_* = \frac{p}{M_H}$, we can obtain that:

$$1 - f_{H*} = \frac{q}{M_T}$$

And further expand $G_{f_{H*}}$ as follows:

$$G_{f_{H*}} = p \ln \frac{p}{M_H} + q \ln \frac{q}{M_T}$$

If, as we said, the Big Q is M_H (and M_T), and big P is p (and q), then we have:

$$G_{f_{H*}} = p \ln \frac{p}{M_H} + q \ln \frac{q}{M_T} = \mathbf{KL_{P||Q}}$$

Thus we have shown that the *Kelly Criterion*—fraction of wealth to wager per bet—maximizes what is equivalent to the KL divergence of the *payoff* distribution with respect to the *true* governing distribution.

Let's finally see that the optimal *Kelly* fractions and wealth growth mirrors what we found for *heads* and *tails*—namely that we should bet *tails*!

$$f_{T*} = 0.5 - \frac{0.5}{0.67} = -0.25$$

$$f_{T*} = 0.5 - \frac{0.5}{1.5} = 0.167$$

Now we plug into G_{f_*} :

$$G_{f_{H*}} = 0.5\ln(1 + 0.67 \times -0.25) + 0.5\ln(1 - -0.25) = 0.0204$$
$$G_{f_{T*}} = 0.5\ln(1 + 1.5 \times 0.167) + 0.5\ln(1 - 0.167) = 0.0204$$

Either f_* will give us the same correct answer! The unified KL form also agrees:

$$G_{f_*} = 0.5\ln\frac{0.5}{0.6} + 0.5\ln\frac{0.5}{0.4} = 0.0204$$

Can we bridge this back to the expected profit values we calculated? Let's try. What exactly is this G_{f_*} value telling us? It leads us to the multiplicative or compound wealth growth rate per bet. Here is the extraction:

$$e^{G_{f*}} - 1 = e^{0.0204} - 1 = 0.0206 = 2.06\,\% \text{ per bet}$$

This tells us that in the long run, betting f_* will grow our wealth at a rate of $2.06\,\%$ per bet. This is different from any expected profit per bet calculation we have done above. For the sake of bridging the gap, though, let's calculate the expected profit per bet using the optimal f_* values.

Once again, we can look at 'shorting' *heads* or riding *tails*:

$$\mathbb{E}_{Profit_{H*}} = f_{H*} \times \mathbb{E}_{Profit_H}$$
$$\mathbb{E}_{Profit_{H*}} = -0.25 \times -0.167$$
$$\mathbb{E}_{Profit_{H*}} = 0.0417 = 4.17\,\% \text{ per bet}$$

And for *tails*:

$$\mathbb{E}_{Profit_{T*}} = f_{T*} \times \mathbb{E}_{Profit_T}$$
$$\mathbb{E}_{Profit_{T*}} = 0.167 \times 0.25$$
$$\mathbb{E}_{Profit_{T*}} = 0.0417 = 4.17\,\% \text{ per bet}$$

Already we can admire the beautiful symmetry that *Kelly* gives us. We can now see that shorting *heads* is precisely the same thing as riding *tails*.

Now, in terms of reconciling this number with the geometric $2.06\,\%$ growth rate, we must make some nuanced points. The $4.17\,\%$ ($\mathbb{E}_{Profit_*}$) rate is the *linear* or *additive* growth rate of the average dollar wagered. This is vitally different in two respects from what $2.06\,\%$, or $e^{G_{f*}}$, is. Not only is $e^{G_{f*}}$ a geometric growth rate, but it is one that explains how our *net wealth* grows. Vitally and definitionally, we are *not* wagering our whole wealth on every bet (and thereby growing geometrically by $4.17\,\%$), but instead are wagering $f_* \times Wealth$ on each bet. Every time we wager this, our wealth grows additively (or is subtracted), in the long run being able to be expressed as an average growth rate. The following Python code will make the G_{f_*} and KL equivalence clear and will reconcile these two different rates.

```
import math
import random

pH = 0.5
pT = 1-pH

qH = 0.6
qT = 0.4

bH = (1-qH)/qH
bT = (1-qT)/qT

W = 1000

# optimal wager fraction
fT = pH - (pT/bT)

Wns = []
for trial in range(10000):
    w = W
    for bet in range(100):
        t = random.random() > 0.5
       # wealth grows according to coin flip
       # on average, this will be an addition of fT * 0.417
        w += fT * w * bT if t else -fT * w
    Wns.append(w)

# take geometric mean to derive G
G = sum(math.log(w/W) for w in Wns) / len(Wns)
print("Estimated G:", G)
```

This code outputs the following:

```
Estimated G: 2.0570345047614063
```

There we are! Every bet, on average, adds $f_* \times \mathbb{E}_{Profit_{T*}}$ to our wealth. When we do this many many times, we see that the average multiplicative growth of *wealth* was just what G_{f_*} predicted: $2.06\,\%$!

Not only have we shown that the G_{f_*} equation, whose maximization gives the optimal betting strategy, is identical to the KL divergence of the *payoff* distribution from the *true* distribution. We have also shown that discrete reasoning about expected additive profits per bet lines up perfectly with long term geometric growth of a portfolio *that performs those discrete bets*.

Returning to Black Scholes and Entropy

Having wet our feet with the provable equivalence of KL divergence (derived from *cross entropy*) and Kelly Wealth Growth, we can return to reasoning about what equivalences there may be between the Black Scholes *implied volatility* and Shannon's *entropy*. Whereas KL tells us about the present divergence of one distribution from another—the information edge—the Black Scholes formula is defined as a constraint requiring *no* informational edge. But this can just as easily then *reveal* the entropies involved in the system that we are hoping to cancel out.

Applying this to our betting example, the Black Scholes does something equivalent to telling the house (the guy offering the bet) exactly what payoff should be offered so as to allow *zero* edge on the part of the bettor. In this case, the answer would be boring because the coin is fair. But if the coin *really did* land *heads* 60 % of the time, the house would need to price it nontrivially to keep itself hedged. Embedded in that calculation is the entropy of the system. This is the constraint:

$$G_{f*} = p \ln(1 + bf_*) + q \ln(1 - f_*) = 0$$

And it will demand a *b* of \$0.67 (and \$1.5, complementarily, for tails).

We will not dive into the mathematics of the Black Scholes equation (here, at least), but I want to reason through why *implied volatility* (IV) is similar to *entropy* (H), now that we have seen how probabilistic financial edge and informational divergence can represent the same thing. In our coins, the edge came from a difference in entropy (a non-zero KL). In financial derivatives, the edge (to be held at 0) is likewise dependent on the difference in entropy of a financial asset (like a stock) and a risk free bond (which has definitionally 0 entropy in returns).

Let's think about time scale, continuity, units, and context. In variable length binary encodings, the entropy tells us the average amount of bits needed to represent *one* discrete event in a stream of *N* discrete events that follow a uniform distribution. We will see many discrete words/units that need encoding and rarely will *any* of them have H bits in their codeword. It is an average that compresses the nature of the distribution into one value. We could also then imagine the average number of bits we'd need to encode, say, 10 word sentences. Since H is already in logarithmic space, then 10 words with entropy H will actually have a net entropy (average amount of bits for whole sentence) will be $H_{10} = \mathrm{H} \times 10$.

This brings us to IV. For simplicity, we can imagine that every day, a stock moves by a discrete amount and we can forget about risk free rate (imagine it is 0.00 %). In this way, daily prices are like *words* and multi-day option expiration schedules are like *sentences*. IV tells us the standard daily deviation of log returns, and thus inform our estimate for average returns over the course of an entire sequence of these discrete changes. Black Scholes price for an option lasting 10 would be a value like IV_{10}. However, since it is a measure of standard deviation, IV *does not* scale additively when repeatedly applied (specifically it scales by $\sqrt{time}$).[1] If we want to align with entropy, we want to *square* IV to get variance. IV^2, then, is analogous to H, and $IV_{10}^2 = IV^2 \times 10$.

And indeed, in the Black Scholes Option Pricing model, we have the term $\sigma^2 T$, where σ is exactly our IV, and *T* is the number of days remaining (like the 10 days in our imagined contract). I am

[1] Suppose that on a given day, the stock can either go up or down by exactly 1.0 % . The one day mean of log returns is 0.0 % , and the variance is $0.5 \times (1.0 - 0.0)^2 + 0.5 \times (-1.0 - 0.0)^2 = 0.5 + 0.5 = 1.0$. Standard deviation is $\sqrt{1.0} = 1.0$. For two days, we still have a mean of 0.0, but our variance becomes $0.25 \times 2.0^2 + 0.5 \times 0.0 + 0.25 \times (-2.0)^2 = 0.25 \times 4.0 + 0.25 \times 4.0 = 2.0$ and the standard deviation is $\sqrt{2.0} = 1.414$. This is a contrived and simplified example, but we see that it is *variance* which is additive with time, as opposed to standard deviation which grows with the *square root* of time.

ignoring many vital details of the Black Scholes price, but the key beauty I am gesturing at is that IV^2, or *implied variance,* is directly analogous to Shannon's Entropy.